

DOMANDE D'ESAME

TARGETSCAN VS MIRTARBASE

DIFFERENZA TRA MICRORNA E UN ALTRO GENE. MIRNA CODIFICA UNA PROTEINA?
SPIEGARE PASSO PASSO IL BOX PLOT. IQR ALTO $\rightarrow$ ALTA VARIANZA

COSA È PUBMED?

CHE CLUSTERING HAI USATO? GERARCHICO? QUALE MODO?
PERCHÉ SI AGGIUSTA IL P-VALUE?

COSA È UNA DISTRIBUZIONE NORMALE?

MOLECOLA DEL DNA RISPETTO A QUELLA DELL'RNA

COSA È ENRICHMENT ANALYSIS? QUALE TEST USA?

COSA È IL χ -TEST?

$\hookrightarrow$ TEST IPERGEOMETRICO

RATIO DEL TEST IPERGEOMETRICO

GENEONTOLOGY DEFINIZIONE. COM'È STRUTTURATO?

DIFFERENZA TRA PATHWAY (COME LA DISEGNI? FLUSSO) E NON HO SENTITO

KEGG

KEGG

KEGG è un database di pathway biologici e processi cellulari.
Serve per capire in quali funzioni biologiche partecipano i geni.

Esempi:

- pathway del cancro
- metabolismo
- signaling pathways
- risposta immunitaria

Nel tuo progetto lo usi nell'**enrichment analysis** per vedere se i DEG appartengono a specifiche vie biologiche.

Quindi KEGG risponde a:

“Questi geni lavorano insieme in quale processo biologico?”

NCBI

NCBI

NCBI è il grande portale centrale di bioinformatica del NIH americano.
Contiene tantissimi database biologici.

Include:

- Gene
- PubMed
- RefSeq
- BLAST
- Genome
- ClinVar
- dbSNP

È più una “piattaforma madre”.

Risponde a:

“Dove trovo informazioni ufficiali su geni, sequenze, mutazioni e letteratura?”



HPA

Human Protein Atlas

HPA = Human Protein Atlas.

Mostra:

- dove un gene/proteina è espresso
- in quali tessuti
- in quali tumori
- immagini istologiche reali

Molto utile per validare biomarker.

Nel tuo caso:

- puoi verificare se FOXJ1 o SFTPC sono alti nel polmone normale e persi in LUSC.

Risponde a:

“Dove viene espressa questa proteina nel corpo umano?”

RefSeq

RefSeq

RefSeq è un database di sequenze di riferimento curate.

Contiene:

- DNA
- mRNA
- proteine

È usato come standard ufficiale per identificare geni e trascritti.

Per esempio:

- un gene può avere diversi isoform;
- RefSeq definisce le sequenze “canoniche”.

Risponde a:

“Qual è la sequenza di riferimento ufficiale di questo gene?”

miRTarBase

miRTarBase

Database di interazioni validate tra:

- miRNA
- geni target

La cosa importante è che le interazioni sono:

- sperimentalmente validate
- non solo predette

Quindi è molto affidabile.

Nel tuo progetto serve per:

- capire quali miRNA regolano i DEG.

Risponde a:

“Quali miRNA controllano realmente questo gene?”

BLAST

BLAST

BLAST confronta sequenze biologiche.

Tu inserisci:

- una sequenza DNA/proteina

e BLAST cerca:

- sequenze simili in database biologici.

Serve per:

- identificare geni sconosciuti
- trovare omologie
- confrontare specie

Risponde a:

“Questa sequenza a cosa assomiglia?”

TargetScan

TargetScan

TargetScan predice target di miRNA usando algoritmi computazionali.

Diversenza fondamentale:

- miRTarBase = interazioni validate sperimentalmente
- TargetScan = predizioni bioinformatiche

Quindi TargetScan è più ampio ma meno certo.

Risponde a:

“Quali geni potrebbe regolare questo miRNA?”

Differenze principali

Database/Tool	Serve per	Tipo
KEGG	Pathway biologici	Functional analysis
NCBI	Portale generale bioinformatico	Database hub
HPA	Espressione proteica/tessuti	Expression atlas
RefSeq	Sequenze ufficiali	Reference sequences
miRTarBase	miRNA-target validati	Experimental interactions
BLAST	Confronto sequenze	Alignment tool
TargetScan	Predizione target miRNA	Predictive algorithm

“What is a gene?”

“A gene is a DNA sequence containing the instructions to produce a functional molecule, usually a protein or a functional RNA.
Genes regulate cellular functions such as proliferation, metabolism, signaling, and differentiation.”

“What is gene expression?”

“Gene expression represents how active a gene is inside a cell.
In RNA-seq analysis, expression is measured by quantifying RNA transcripts produced from each gene.”

“What are DEGs?”

“DEGs are Differentially Expressed Genes, meaning genes whose expression changes significantly between tumor and normal tissues.”

“Why do we calculate DEGs?”

“Because they help identify genes potentially involved in tumor development, progression, or loss of normal cellular functions.”

“What is log Fold Change?”

“Log Fold Change measures how much a gene changes between two conditions.
Positive values indicate upregulation in tumors, while negative values indicate downregulation.”

“Why use log2 Fold Change instead of normal fold change?”

“The logarithmic scale makes upregulation and downregulation symmetric and easier to visualize and interpret statistically.”

“Why did you choose fold change = 2.5?”

“To focus only on genes with strong biological differences between tumor and normal tissues, reducing weak or noisy variations.”

“What is a p-value?”

“The p-value estimates the probability that the observed difference happened by chance.”

“Why do we adjust the p-value?”

“Because thousands of genes are tested simultaneously.
Without correction, many genes would appear significant just by random chance.”

“What is FDR?”

“FDR stands for False Discovery Rate.
It controls the expected proportion of false positives among significant genes.”

“What is IQR?”

“IQR, or Interquartile Range, measures the variability of gene expression values.

Genes with very low IQR show almost no variation across samples and are usually biologically uninformative.”

“Why remove low-IQR genes?”

“Because non-variable genes mainly add noise and reduce the statistical power of downstream analyses.”

“Why use matched tumor-normal pairs?”

“Matched samples reduce patient-to-patient variability because tumor and normal tissues come from the same individual.”

“Why use TCGA?”

“TCGA is a large public cancer genomics database containing high-quality RNA-seq and clinical data from many cancer types, including LUSC.”

“What does the volcano plot show?”

“The volcano plot combines magnitude of change and statistical significance. It allows rapid identification of strongly deregulated genes.”

“Why are up genes on the right and down genes on the left?”

“Because positive log fold change values indicate higher expression in tumors, while negative values indicate lower expression.”

“What does the heatmap show?”

“The heatmap visualizes expression patterns across samples and genes. It helps identify clustering and separation between tumor and normal tissues.”

“What is hierarchical clustering?”

“It is an unsupervised method that groups samples with similar expression profiles.”

“Why is the heatmap important?”

“Because it demonstrates that tumor and normal tissues have clearly distinct transcriptional profiles.”

“What does upregulated mean?”

“The gene is expressed more in tumor tissues compared to normal tissues.”

“What does downregulated mean?”

“The gene is expressed less in tumor tissues compared to normal tissues.”



“Why are there more downregulated genes than upregulated genes?”

“This suggests that LUSC not only activates oncogenic pathways, but also suppresses many normal lung and immune-related functions.”

“What is RNA-seq?”

“RNA sequencing is a technique used to measure transcript abundance and analyze gene expression genome-wide.”

“Why did you use adjusted p-value < 0.05?”

“It is a standard statistical threshold used to reduce false positive discoveries while maintaining sensitivity.”

“What does the boxplot notch mean?”

“The notch represents an approximate confidence interval around the median. If notches do not overlap, the medians are likely significantly different.”

“Why use paired t-test later?”

“Because tumor and normal samples come from the same patients, so paired testing better accounts for inter-patient variability.”

What is enrichment analysis?

Enrichment analysis is a statistical method used to identify biological functions, pathways, or diseases that are overrepresented in a list of genes compared to what would be expected by chance.

In your case, it was used to understand the biological meaning of the DEG identified in LUSC.

Why do we perform enrichment analysis?

Because a list of hundreds of DEG alone is difficult to interpret biologically.

Enrichment analysis helps transform gene lists into:

- biological pathways,
 - cellular processes,
 - molecular functions,
 - disease associations.
-

Which statistical test does EnrichR mainly use?

EnrichR mainly uses the **hypergeometric test** to calculate enrichment significance.

Then it adjusts p-values for multiple testing using FDR correction.

What is the hypergeometric test?

The hypergeometric test evaluates whether the overlap between:

- your DEG list,
- and a biological pathway/gene set

is larger than expected by random chance.

It answers:

“Is this pathway enriched in my genes more than expected randomly?”

What does the hypergeometric test compare?

It compares:

- total genes in the genome,
 - genes in your DEG list,
 - genes belonging to a pathway,
 - overlapping genes between the two sets.
-

What is gene ratio in enrichment plots?

Gene ratio indicates:

$$\text{Gene Ratio} = \frac{\text{Number of DEG in pathway}}{\text{Total genes in pathway}}$$

Higher ratio means a larger proportion of pathway genes are differentially expressed.

What does adjusted p-value mean in enrichment analysis?

Many pathways are tested simultaneously.

Adjusted p-value (FDR) corrects for multiple testing to reduce false positives.

Smaller adjusted p-values indicate more reliable enrichment.

Why is FDR correction necessary?

Because testing hundreds or thousands of pathways increases the probability of obtaining significant results by chance.

FDR controls this problem.

What is GO (Gene Ontology)?

Gene Ontology is a structured biological annotation system used to describe gene functions consistently across species.

How is Gene Ontology structured?

GO has three main categories:

- 1. GO Biological Process (BP)**
Biological programs and processes
Example: cell cycle, immune response
 - 2. GO Molecular Function (MF)**
Molecular activities of proteins
Example: DNA binding, cytokine activity
 - 3. GO Cellular Component (CC)**
Cellular localization
Example: nucleus, membrane, ribosome
-

Why did you use GO Biological Process?

To identify altered biological programs active in LUSC, such as:

- mitosis,
 - immune regulation,
 - DNA replication.
-

Why did you use GO Molecular Function?

To study altered molecular activities of genes and proteins.

What is KEGG?

KEGG is a pathway database that maps genes into known signaling and metabolic pathways.

Example:

- Cell cycle,
 - p53 signaling,
 - antigen presentation.
-

Why did you use KEGG?

Because it helps visualize coordinated pathway alterations rather than isolated genes.

What is DisGeNET?

DisGeNET is a database linking genes to diseases.

It helps identify disease associations enriched in the DEG list.

What is a pathway?

A pathway is a set of interacting genes/proteins involved in a biological process.

Example:

- cell cycle pathway,
 - antigen presentation pathway.
-

Why are cell cycle pathways enriched in tumors?

Cancer cells proliferate rapidly, requiring activation of:

- mitosis,
 - DNA replication,
 - checkpoint regulation.
-

Why are immune pathways downregulated in LUSC?

Tumors often suppress immune functions to evade immune recognition and destruction.

What is antigen presentation?

It is the process by which cells expose antigen peptides on MHC molecules to activate T cells.

Why is antigen presentation important in cancer?

Reduced antigen presentation helps tumors escape immune surveillance.

What is the difference between GO and KEGG?

- **GO** describes gene functions and biological roles.
- **KEGG** describes interactions between genes inside pathways.

Example:

- GO → "mitotic spindle organization"
 - KEGG → "cell cycle pathway"
-

Why use multiple enrichment databases instead of only one?

Because different databases provide complementary biological interpretations:

- GO → functions/processes
- KEGG → pathways
- DisGeNET → disease associations

Using multiple sources increases biological reliability.

Why did you perform enrichment separately for upregulated and downregulated genes?

Because activated and suppressed genes often belong to completely different biological programs.

Combining them could hide important patterns.

What is cytokine signaling?

Cytokines are signaling molecules used by immune cells to communicate and coordinate inflammatory responses.

What is mitosis?

Mitosis is the process through which a cell divides into two daughter cells.

What is DNA replication?

It is the duplication of DNA before cell division.

Why is DNA replication enrichment important in tumors?

Because rapidly proliferating tumor cells require continuous DNA synthesis.

What is a normal distribution?

A normal distribution is a symmetric bell-shaped distribution where most values are concentrated around the mean, while extreme values are progressively rarer.

Main properties:

- mean $\approx$ median $\approx$ mode
- symmetric shape
- many statistical tests assume normality

Example:

- human height,
 - measurement errors,
 - many biological variables approximately follow it.
-

Why is normal distribution important in statistics?

Because many statistical tests, such as:

- t-test,
- ANOVA,
- linear regression

assume that data are approximately normally distributed.

What happens if data are not normally distributed?

Parametric tests may become unreliable.

In those cases:

- data transformation,
- or non-parametric tests

may be preferred.

What is the difference between DNA and RNA?

DNA and RNA are both nucleic acids, but they have different roles.

DNA

- stores genetic information
- double-stranded
- contains deoxyribose sugar
- uses thymine (T)

RNA

- transfers or regulates genetic information
 - usually single-stranded
 - contains ribose sugar
 - uses uracil (U) instead of thymine
-

What is the main function of DNA?

DNA stores the genetic instructions required for cell function and inheritance.

What is the main function of RNA?

RNA helps convert genetic information into functional molecules and regulates gene expression.

What are the main types of RNA?

- mRNA → messenger RNA
- tRNA → transfer RNA
- rRNA → ribosomal RNA
- miRNA → microRNA

What is mRNA?

Messenger RNA carries genetic information from DNA to ribosomes for protein synthesis.

What is a microRNA?

A microRNA (miRNA) is a small non-coding RNA molecule that regulates gene expression post-transcriptionally.

Does a microRNA encode proteins?

No.

miRNAs are **non-coding RNAs**, meaning they do not produce proteins.

Their role is regulatory.

How does a microRNA work?

miRNAs bind complementary sequences on target mRNAs and:

- block translation,
- or promote mRNA degradation.

As a result, protein production decreases.

Why are miRNAs important in cancer?

Because they regulate many genes involved in:

- proliferation,
 - apoptosis,
 - invasion,
 - metastasis,
 - immune response.
-

What is a tumor suppressor miRNA?

A tumor suppressor miRNA inhibits oncogenic processes.

If its activity decreases:

- oncogenes may become overexpressed,
 - tumor progression may increase.
-

What is an oncogenic miRNA?

An oncogenic miRNA ("oncomiR") promotes tumor development by suppressing tumor suppressor genes.

Why can reduced miRNA activity increase tumor growth?

Because the genes normally controlled by the miRNA lose regulation and become overexpressed.

What does "post-transcriptional regulation" mean?

It means regulation occurring after mRNA has already been transcribed from DNA.

miRNAs mainly act at this level.

Cos'è un biomarker?

Un biomarker è una caratteristica biologica misurabile che indica uno stato normale o patologico.

Nel tuo caso, i biomarker sono geni la cui espressione cambia significativamente tra tessuto normale e tumorale.

Possono essere usati per:

- diagnosticare una malattia;
- prevedere la prognosi;
- monitorare la progressione tumorale;
- predire la risposta a una terapia.

Esempio:

- **CDC20** → biomarker proliferativo (tumore aggressivo);
 - **SFTPC** → biomarker della perdita dell'identità polmonare normale.
-

Cos'è un miRNA?

I miRNA (microRNA) sono piccoli RNA non codificanti, lunghi circa 20–22 nucleotidi, che regolano l'espressione genica a livello post-trascrizionale.

Non codificano proteine.

Il loro compito è legarsi agli mRNA bersaglio e:

- degradarli,
- oppure bloccarne la traduzione.

Quindi funzionano come "freni" dell'espressione genica.

Nel tumore, la loro alterazione può causare:

- attivazione di oncogeni,
 - perdita di tumor suppressor,
 - aumento proliferazione e metastasi.
-

Cos'è MIENTURNET?

MIENTURNET è un tool bioinformatico per analizzare reti miRNA-gene.

Prende una lista di geni e:

1. identifica i miRNA che li regolano;
2. costruisce network di interazioni;
3. esegue enrichment analysis sui miRNA.

Nel tuo progetto lo hai usato per:

- trovare miRNA associati ai DEG;
 - identificare i principali regolatori;
 - costruire le network miRNA-target.
-

Domanda difficile che potrebbero farti

“Perché un tumor suppressor miRNA compare associato ai geni upregolati?”

Perché:

- il miRNA potrebbe essere ridotto o non funzionare correttamente;
- quindi perde la capacità di reprimere i suoi target oncogenici;
- di conseguenza quei geni diventano upregolati nel tumore.

È esattamente il caso discusso per miR-193b-3p.

Quali sono i nodi di GO?

GO (*Gene Ontology*) è organizzato come una rete gerarchica di termini biologici.

I “nodi” di GO sono:

termini biologici chiamati **GO terms**.

Sono divisi in 3 categorie principali:

- **Biological Process (BP)**
→ processi biologici
es. cell cycle, apoptosis
 - **Molecular Function (MF)**
→ funzione molecolare di una proteina
es. ATP binding
 - **Cellular Component (CC)**
→ dove agisce nella cellula
es. nucleus, mitochondrion
-

Cosa è un pathway?

Un pathway è:

una serie organizzata di interazioni biologiche che portano a una funzione cellulare.

Esempi:

- cell cycle
- DNA replication
- apoptosis
- antigen presentation

Nei pathway:

- geni/proteine lavorano insieme,
- ogni elemento ha un ruolo specifico.

K-means clustering vs Hierarchical clustering

K-means	Hierarchical
Divide i dati in K gruppi	Costruisce un albero (dendrogramma)
Devi scegliere K prima	Non serve scegliere K
Basato sulla distanza dal centroide	Basato sulla distanza tra campioni
Più veloce	Più interpretabile
Usato per grandi dataset	Molto usato nelle heatmap

Nel tuo progetto hai usato:

Hierarchical clustering nella heatmap.

Cosa è IQR? E nel boxplot?

$IQR = \text{Interquartile Range}$

$$IQR = Q3 - Q1$$

Misura la dispersione del 50% centrale dei dati.

Nel boxplot:

- il box va da Q1 a Q3,
- quindi l'altezza del box = IQR.

Più il box è alto:

→ più i dati sono variabili.

Cosa sono i gene expression data?

Sono dati che misurano:

quanto un gene è espresso in un campione.

In pratica:

- quanta RNA produce un gene.

Dove si trovano?

Database pubblici:

- TCGA
- GEO
- GTEx
- ArrayExpress

Tu hai usato:

TCGA-LUSC.

Differenza tra gene expression data e DNA sequence

DNA sequence

Gene expression

Sequenza genetica

Attività dei geni

Quasi uguale in tutte le cellule

Cambia tra tessuti e malattie

Dice “cosa può fare”

Dice “cosa sta facendo”

A,T,C,G

livelli numerici di espressione

Cosa è un percentile?

Un percentile indica:

la percentuale di dati sotto un certo valore.

Esempio:

- 90° percentile
→ il 90% dei dati è sotto quel valore.
-

Cosa è un quartile?

I quartili dividono i dati in 4 parti.

- Q1 = 25%
 - Q2 = mediana = 50%
 - Q3 = 75%
-

Processo da mRNA a proteina

1. Trascrizione

DNA → mRNA

2. Maturazione

splicing, capping, poly-A tail

3. Traduzione

mRNA → proteina (nei ribosomi)

Tutto l'mRNA diventa proteina?

No.

Alcuni mRNA:

- vengono degradati,
- regolati dai miRNA,
- oppure non tradotti.

Inoltre:

- esistono RNA non codificanti,
 - come i miRNA.
-

Cosa è FDR?

FDR = *False Discovery Rate*

Serve per correggere i p-value quando fai molti test contemporaneamente.

Nel tuo progetto:

- hai testato migliaia di geni,
 - senza correzione avresti molti falsi positivi.
-

Idea intuitiva

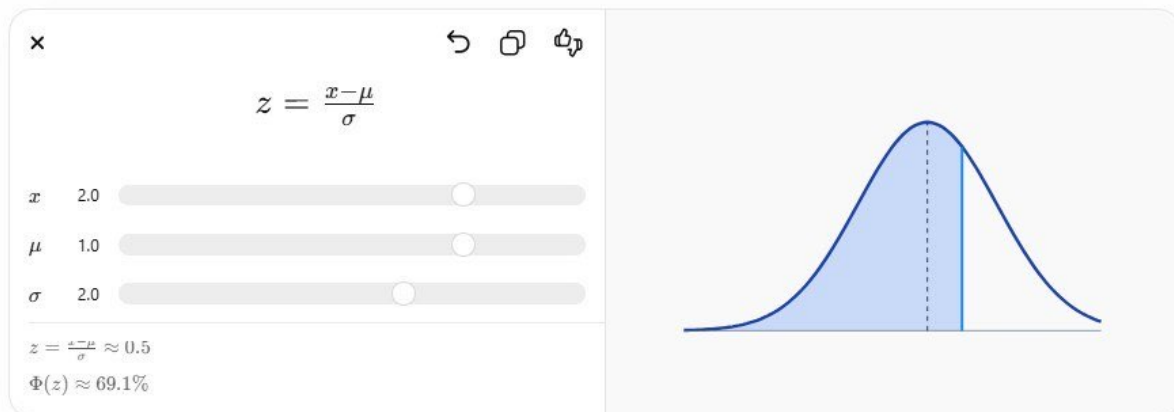
Se $FDR = 0.05$:

circa il 5% dei risultati significativi potrebbe essere falso positivo.

Cosa è lo Z-score?

Lo Z-score misura:

quanto un valore si allontana dalla media.



dove:

- x = valore
- μ = media
- σ = deviazione standard

Interpretazione

Z-score	Significato
0	valore medio
+2	sopra la media
-2	sotto la media